Complete AI / ML / Python / DSA Extended Answer Bank

Python Answers

Basic Questions

1. What is Python and why is it popular?

Python is a high-level programming language known for readability, simplicity, huge community support, and powerful libraries for AI, ML, web development, automation, and data science.

2. Difference between local and global variables.

- Local variables exist inside a function.
- Global variables exist throughout the program.

3. What is type casting?

Converting one data type into another. Example:

```
x = int("5")
```

4. Explain Python indentation.

Indentation defines code blocks in Python instead of curly braces.

5. What is the use of pass statement?

`pass` acts as a placeholder when no code is written yet.

6. Difference between remove() and pop().

- `remove()` deletes by value.
- `pop()` deletes by index.

7. Explain dictionary in Python.

Dictionary stores data as key-value pairs.

8. What is unpacking?

Extracting elements into variables.

```
a,b = [1,2]
```

9. Explain f-strings.

Formatted strings introduced in Python 3.6.

```
name = "Kamal"
print(f"Hello {name}")
```

10. Difference between set and list.

- Set stores unique values.
- List allows duplicates.

Advanced Questions

1. Explain monkey patching.

Changing class or module behavior at runtime.

2. What are Python descriptors?

Objects controlling attribute access using `__get__`, `__set__`, and `__delete__`.

3. Difference between staticmethod and classmethod.

- `staticmethod` does not access class.
- `classmethod` accesses class using `cls`.

4. Explain duck typing.

Object type matters less than behavior. “If it walks like a duck and quacks like a duck, it’s a duck.”

5. What is memoization?

Caching previous function results for faster execution.

6. Explain MRO.

Method Resolution Order determines order of method lookup in inheritance.

7. Difference between threading and asyncio.

- Threading uses multiple threads.
- asyncio uses asynchronous event loop.

8. Explain closures in Python.

Functions remembering variables from enclosing scope.

9. What is serialization and pickling?

Converting object into storable format. Pickle serializes Python objects.

10. Explain dependency injection.

Providing dependencies externally instead of creating internally.

Debugging Answers

Debug 1

```
a = [1,2,3]
print(a*2)
```

Answer: It repeats the list instead of mathematical multiplication. Output:

```
[1,2,3,1,2,3]
```

Debug 2

```
x = "10"
y = 5
print(x+y)
```

Answer: String and integer cannot be added directly. Fix:

```
print(int(x)+y)
```

Debug 3

```
def func():
    try:
        return 1
    finally:
        return 2
```

Answer: Output is `2` because finally block overrides try return.

DSA Answers

Basic Questions

1. What is linear search?

Searching elements one by one sequentially.

2. Difference between stack overflow and underflow.

- Overflow: adding beyond capacity.
- Underflow: removing from empty stack.

3. Explain adjacency matrix.

A 2D matrix representation of graph connections.

4. What is heap data structure?

A complete binary tree used in priority queues.

5. What is collision in hashing?

When two keys generate same hash index.

Advanced Questions

1. Explain Kadane’s algorithm.

Efficient algorithm to find maximum subarray sum.

2. Difference between min heap and max heap.

- Min heap: smallest element at root.
- Max heap: largest element at root.

3. Explain trie data structure.

Tree structure used for efficient string searching.

4. What is topological sorting?

Linear ordering of DAG vertices.

5. Explain union-find algorithm.

Used for disjoint set operations and cycle detection.

Debugging

Debug 1

```
queue = []
queue.pop(0)
```

Answer: Raises IndexError because queue is empty.

Debug 2

```
def factorial(n):
    return n * factorial(n-1)
```

Answer: Missing base condition. Fix:

```
if n == 0:
    return 1
```

EDA / Data Analytics Answers

Basic Questions

1. What is data cleaning?

Fixing missing, incorrect, duplicate, or inconsistent data.

2. Explain categorical and numerical data.

- Categorical: labels/categories.
- Numerical: measurable values.

3. What is data wrangling?

Transforming raw data into useful format.

4. Difference between histogram and bar chart.

- Histogram for continuous data.
- Bar chart for categorical data.

5. Explain percentile.

Value below which a percentage of observations fall.

Advanced Questions

1. Explain PCA.

Principal Component Analysis reduces dimensions while preserving variance.

2. What is feature scaling?

Normalizing feature ranges.

3. Difference between OLAP and OLTP.

- OLAP for analysis.
- OLTP for transactions.

4. Explain one-hot encoding.

Converting categories into binary columns.

5. What is dimensionality reduction?

Reducing number of features while preserving important information.

Debugging

Debug 1

```
df['age'].fillna(df['age'].mean)
```

Answer: Result not assigned back. Fix:

```
df['age'] = df['age'].fillna(df['age'].mean())
```

Debug 2

```
plt.plot(df)
```

Answer: Graph may be unclear due to multiple columns and missing labels.

Machine Learning Answers

Basic Questions

1. What is feature selection?

Selecting important features for better model performance.

2. Explain label encoding.

Converting categories into numerical labels.

3. Difference between batch and online learning.

- Batch learning trains on entire dataset.
- Online learning updates continuously.

4. What is KNN?

Algorithm predicting using nearest neighbors.

5. What is linear regression?

Algorithm predicting continuous values using linear relationship.

Advanced Questions

1. Explain XGBoost.

Efficient boosting algorithm improving weak learners.

2. Difference between ROC and AUC.

- ROC is curve.
- AUC is area under ROC curve.

3. Explain hyperparameter tuning.

Optimizing model settings.

4. What is cross-validation?

Technique evaluating model across multiple splits.

5. Explain ensemble learning.

Combining multiple models for better performance.

Debugging

Debug 1

```
model.predict(y_test)
```

Answer: `predict()` expects features not labels.

Debug 2

```
model = LogisticRegression
```

Answer: Missing parentheses. Correct:

```
model = LogisticRegression()
```

Deep Learning Answers

Basic Questions

1. What is perceptron?

Simplest neural network unit.

2. Explain hidden layers.

Intermediate layers learning patterns.

3. What is loss function?

Measures prediction error.

4. Difference between sigmoid and ReLU.

- Sigmoid outputs 0–1.
- ReLU outputs positive values.

5. Explain tensor.

Multi-dimensional data structure used in DL.

Advanced Questions

1. Explain batch normalization.

Normalizes activations for stable training.

2. What is exploding gradient problem?

Gradients become excessively large during training.

3. Explain attention mechanism.

Model focuses on important parts of input.

4. What is transformer architecture?

Deep learning architecture using self-attention.

5. Explain GANs.

Generative Adversarial Networks use generator and discriminator.

Debugging

Debug 1

```
model.fit(X_train)
```

Answer: Missing target labels `y_train`.

Debug 2

```
Dense(10, activation='relu')
```

Answer: For classification final layer often needs sigmoid or softmax.

Scenario-Based Answers

Python

Script works locally but fails on server.

Check:

- Dependency versions

- Environment variables
- Logs
- File paths
- Python version

Program consuming too much RAM.

Use:

- Profiling
- Generators
- Garbage collection
- Efficient structures

DSA Scenario Answers

Instant search with millions of records.

Use Hash Tables or Trie.

Real-time navigation algorithm.

Use Dijkstra or A* algorithm.

EDA Scenario Answers

Dataset has 90% duplicate rows.

Investigate source then remove duplicates carefully.

Negative age values.

Treat as invalid data and clean/impute.

ML Scenario Answers

Accuracy dropped after deployment.

Possible:

- Data drift
- Overfitting
- Changed real-world patterns

Explain ML prediction to non-technical client.

Use simple examples and feature importance.

DL Scenario Answers

CNN performs poorly on new images.

Possible:

- Overfitting
- Poor augmentation
- Low-quality training data

Training takes 15 hours.

Optimize using:

- GPU
- Batch tuning
- Smaller model
- Mixed precision

Rapid Coding Challenge Concepts

Reverse words in sentence.

Use split and reverse.

Merge two sorted arrays.

Use two-pointer technique.

Find common elements.

Use set intersection.

Queue using stacks.

Use two stacks.

Detect cycle in linked list.

Use Floyd’s cycle detection.

Find majority element.

Use Boyer-Moore Voting Algorithm.

Recommendation system logic.

Use similarity-based recommendations.

Chatbot using if-else.

Use conditional responses.

Spam detector logic.

Use keywords or ML classifier.

URL shortener logic.

Generate unique hash key.

HR + Technical Answers

Explain your favorite project.

Describe objective, technologies, challenges, and outcomes.

Biggest bug faced.

Explain issue, debugging process, and learning.

Learning new technologies.

Use documentation, projects, practice, and tutorials.

Explain project failure.

Discuss failure honestly and lessons learned.

Handling deadlines.

Prioritize tasks and manage time effectively.

Mini Viva Answers

Why use Python for AI?

Due to simplicity and AI libraries.

Why use NumPy instead of lists?

Faster numerical computation.

Difference between fit() and transform().

- fit() learns.
- transform() applies.

Why use GPUs in deep learning?

Parallel processing speeds matrix operations.

What is data leakage?

Using future/test information during training.

What is activation function?

Adds non-linearity.

Difference between AI and automation.

- Automation follows rules.
- AI learns patterns.

Explain confusion matrix.

Table showing prediction performance.

What is recursion base case?

Stopping condition for recursion.

Why is clean data important?

Better data produces better models.